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Mathematical Foundations of Data Science (Probability)

Lecture 02

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Probability Measure

Fix a measurable space $(\Omega, \mathcal{F})$.

Definition (Probability Measure)

A function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is called a **probability measure** if the following properties are satisfied:

1. $\mathbb{P}(\emptyset) = 0$.
2. $\mathbb{P}(\Omega) = 1$.
3. If $A_1, A_2, \dots$ is a countably infinite collection of **mutually disjoint** sets, with $A_k \in \mathcal{F}$ for each $k \in \mathbb{N}$ and $A_k \cap A_\ell = \emptyset$ for all $k \neq \ell$, then

$$\mathbb{P} \left(\bigsqcup_{k \in \mathbb{N}} A_k \right) = \sum_{k \in \mathbb{N}} \mathbb{P}(A_k).$$

The triplet $(\Omega, \mathcal{F}, \mathbb{P})$ is called a **probability space**

Properties of Probability Measures

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- For any two **disjoint** sets $A, B \in \mathcal{F}$,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B).$$

- For any $n \in \mathbb{N}$ and a collection of **mutually disjoint** sets $B_1, \dots, B_n \in \mathcal{F}$,

$$\mathbb{P}\left(\bigsqcup_{k=1}^n B_k\right) = \sum_{k=1}^n \mathbb{P}(B_k).$$

- For any set $A \in \mathcal{F}$,

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A).$$

Properties of Probability Measures

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- **(Monotonicity)**

For any two sets $A, B \in \mathcal{F}$ such that $A \subseteq B$,

$$\mathbb{P}(A) \leq \mathbb{P}(B).$$

Equality above holds if and only if $\mathbb{P}(B \setminus A) = 0$

- For any two sets $A, B \in \mathcal{F}$,

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

- **(Inclusion-Exclusion Principle)**

For any $n \in \mathbb{N}$ and any events $A_1, \dots, A_n \in \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \mathbb{P}(A_i) - \sum_{i < j} \mathbb{P}(A_i \cap A_j) + \sum_{i < j < k} \mathbb{P}(A_i \cap A_j \cap A_k) - \dots + (-1)^{n+1} \mathbb{P}\left(\bigcap_{i=1}^n A_i\right).$$

Probability Assignment – Examples

- Suggest a probability assignment $\mathbb{P}$ for

$$\Omega = \{H, T\}, \quad \mathcal{F} = 2^\Omega = \left\{ \emptyset, \Omega, \{H\}, \{T\} \right\}.$$

- Suggest a probability assignment $\mathbb{P}$ for

$$\Omega = \{1, 2, \dots, 6\}, \quad \mathcal{F} = \left\{ \emptyset, \Omega, \{1\}, \{2, 3\}, \right.$$

Probability Assignment – Examples

- Suggest a probability assignment $\mathbb{P}$ for

$$\Omega = \{H, T\}, \quad \mathcal{F} = 2^\Omega = \left\{ \emptyset, \Omega, \{H\}, \{T\} \right\}.$$

- Suggest a probability assignment $\mathbb{P}$ for

$$\Omega = \{1, 2, \dots, 6\}, \quad \mathcal{F} = \left\{ \emptyset, \Omega, \{1\}, \{2, 3\}, \{1, 2, 3\}, \{4, 5, 6\}, \{1, 4, 5, 6\}, \{2, 3, 4, 5, 6\} \right\}.$$

- Suggest a probability assignment $\mathbb{P}$ for

$$\Omega = \{1, \dots, 6\}, \quad \mathcal{F} = \left\{ \emptyset, \Omega, \{1, 2\}, \{3, 4\}, \{5, 6\}, \{1, 2, 3, 4\}, \{3, 4, 5, 6\}, \{1, 2, 5, 6\} \right\}.$$

On Zero-Probability Sets

- $\mathbb{P}(A) = 0 \not\Rightarrow A = \emptyset$
- $\mathbb{P}(A) = 1 \not\Rightarrow A = \Omega$

Continuity of Probability Measure – 1

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- If $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$, where $A_k \in \mathcal{F}$ for each $k \in \mathbb{N}$, then

$$\mathbb{P}\left(\bigcup_{k \in \mathbb{N}} A_k\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n).$$

Proof:

- Let $B_1, B_2, \dots$ be defined as

$$B_1 = A_1, \quad B_2 = A_2 \setminus A_1, \quad B_3 = A_3 \setminus A_2, \quad \dots$$

- **Claim 1:** $B_i \cap B_j = \emptyset$ for all $i \neq j$
- **Claim 2:** We have

$$\bigsqcup_{k=1}^n B_k = \bigcup_{k=1}^n A_k = A_n \quad \forall n \in \mathbb{N}, \quad \bigsqcup_{k \in \mathbb{N}} B_k = \bigcup_{k \in \mathbb{N}} A_k.$$

- Therefore, it follows that

$$\mathbb{P}\left(\bigcup_{k \in \mathbb{N}} A_k\right) = \mathbb{P}\left(\bigsqcup_{k \in \mathbb{N}} B_k\right) = \sum_{k \in \mathbb{N}} \mathbb{P}(B_k) = \lim_{n \rightarrow \infty} \sum_{k=1}^n \mathbb{P}(B_k) = \lim_{n \rightarrow \infty} \mathbb{P}\left(\bigsqcup_{k=1}^n B_k\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n).$$

Continuity of Probability Measure – 2

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- If $A_1 \supseteq A_2 \supseteq A_3 \supseteq \cdots$, where $A_k \in \mathcal{F}$ for each $k \in \mathbb{N}$, then

$$\mathbb{P}\left(\bigcap_{k \in \mathbb{N}} A_k\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n).$$

Proof:

- Follows from the proof of the previous result with $B_k = A_k^c$; the details are left as exercise

Union Bound

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- For any $A_1, A_2, \dots \in \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{k \in \mathbb{N}} A_k\right) \leq \sum_{k \in \mathbb{N}} \mathbb{P}(A_k).$$

Proof:

- Let $B_1, B_2, \dots$ be defined as

$$B_1 = A_1, \quad B_2 = A_2 \setminus A_1, \quad B_3 = A_3 \setminus (A_1 \cup A_2), \quad B_4 = A_4 \setminus (A_1 \cup A_2 \cup A_3), \quad \dots$$

- Claim 1:** $B_i \cap B_j = \emptyset$ for all $i \neq j$
- Claim 2:** We have

$$B_k \subseteq A_k \quad \forall k \in \mathbb{N}, \quad \bigsqcup_{k \in \mathbb{N}} B_k = \bigcup_{k \in \mathbb{N}} A_k.$$

- Therefore, it follows that

$$\mathbb{P}\left(\bigcup_{k \in \mathbb{N}} A_k\right) = \mathbb{P}\left(\bigsqcup_{k \in \mathbb{N}} B_k\right) = \sum_{k \in \mathbb{N}} \mathbb{P}(B_k) \leq \sum_{k \in \mathbb{N}} \mathbb{P}(A_k).$$

Union Bound

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- For any $A_1, A_2, \dots \in \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{k \in \mathbb{N}} A_k\right) \leq \sum_{k \in \mathbb{N}} \mathbb{P}(A_k).$$

Union Bound

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- For any $A_1, A_2, \dots \in \mathcal{F}$,

$$\mathbb{P}\left(\bigcup_{k \in \mathbb{N}} A_k\right) \leq \sum_{k \in \mathbb{N}} \mathbb{P}(A_k).$$

Corollary

For any two sets $A, B \in \mathcal{F}$ (not necessarily disjoint),

$$\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B).$$

More generally, for any $n \in \mathbb{N}$ and sets $A_1, \dots, A_n \in \mathcal{F}$ (not necessarily mutually disjoint),

$$\mathbb{P}\left(\bigcup_{k=1}^n A_k\right) \leq \sum_{k=1}^n \mathbb{P}(A_k).$$

Equality in the Union Bound

Equality in the Union Bound

When does equality hold in the union bound?

- From the proof, it is evident that for equality to hold in the union bound,

$$\mathbb{P}(B_k) = \mathbb{P}(A_k) \quad \forall k \in \mathbb{N}$$

- That is:

$$\begin{aligned} & \mathbb{P}(B_1) = \mathbb{P}(A_1) \\ & \mathbb{P}(B_2) = \mathbb{P}(A_2 \setminus A_1) = \mathbb{P}(A_2) \quad \Longleftrightarrow \quad \mathbb{P}(A_2 \cap A_1) = 0 \\ & \mathbb{P}(B_3) = \mathbb{P}(A_3 \setminus (A_1 \cup A_2)) = \mathbb{P}(A_3) \quad \Longleftrightarrow \quad \mathbb{P}(A_3 \cap (A_1 \cup A_2)) = 0 \end{aligned}$$

For the union bound to hold with equality,

$$\mathbb{P} \left(A_k \cap \left(\bigcup_{j=1}^{k-1} A_j \right) \right) = 0 \quad \forall k > 1.$$

- A **sufficient** condition for equality in the union bound is that $A_1, A_2, \dots$ are mutually disjoint
Mutual disjointness is **not necessary** for equality in the union bound

Example

- Consider the measurable space

$$(\Omega, \mathcal{F}) = (\{1, \dots, 6\}, 2^\Omega)$$

- Consider the events

$$A = \{1, 2, 3, 4\}, \quad B = \{2, 3, 5\}.$$

- Clearly, $A \cap B = \{2, 3\} \neq \emptyset$
- Consider the following probability assignment:

$$\mathbb{P}(\{1\}) = \mathbb{P}(\{4\}) = \mathbb{P}(\{5\}) = \mathbb{P}(\{6\}) = \frac{1}{4}.$$

- Observe that

$$\mathbb{P}(A) = \frac{1}{2}, \quad \mathbb{P}(B) = \frac{1}{4}, \quad \mathbb{P}(A \cup B) = \frac{3}{4}, \quad \mathbb{P}(A \cap B) = 0.$$

Therefore, union bound holds with equality, but A and B are not mutually disjoint

Probability Measures on Discrete Sample Spaces

Discrete Sample Spaces

- Let Ω be a non-empty, **discrete** (finite or countably infinite) sample space
- Then, the elements of Ω may be enumerated as

$$\Omega = \{\omega_1, \omega_2, \dots\}$$

- In this case, we simply take $\mathcal{F} = 2^\Omega$

Probability Assignment for Discrete Sample Spaces

- Specify $\mathbb{P}(\{\omega\})$ for each $\omega \in \Omega$
- For any $A \in \mathcal{F}$:

$$A = \bigsqcup_{\omega \in A} \{\omega\}, \quad \mathbb{P}(A) := \sum_{\omega \in A} \mathbb{P}(\{\omega\})$$

- Different probability assignments $\mathbb{P}$ result from different ways of specifying $\mathbb{P}(\{\omega\})$ for $\omega \in \Omega$

Some Popular Probability Assignments

- $\Omega = \{0, 1\}, \quad \mathcal{F} = 2^\Omega = \left\{ \emptyset, \Omega, \{0\}, \{1\} \right\}$

Fix $p \in [0, 1]$. The measure $\mathbb{P}$ defined by

$$\mathbb{P}(\{1\}) = p, \quad \mathbb{P}(\{0\}) = 1 - p,$$

is popularly called **Bernoulli(p) measure**

- Fix $n \in \mathbb{N}$.
 $\Omega = \{1, \dots, n\}, \quad \mathcal{F} = 2^\Omega$

The measure $\mathbb{P}$ defined by

$$\mathbb{P}(\{k\}) = \frac{1}{n}, \quad k \in \Omega,$$

is popularly called **discrete uniform measure**

Some Popular Probability Assignments

- Fix $n \in \mathbb{N}$.

$$\Omega = \{0, \dots, n\}, \quad \mathcal{F} = 2^\Omega.$$

Fix $p \in [0, 1]$. The measure $\mathbb{P}$ defined by

$$\mathbb{P}(\{k\}) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k \in \Omega,$$

is popularly called **Binomial(n, p) measure**

Some Popular Probability Assignments

- $\Omega = \mathbb{N} = \{1, 2, \dots\}$, $\mathcal{F} = 2^\Omega$.

Fix $p \in [0, 1]$. The measure $\mathbb{P}$ defined by

$$\mathbb{P}(\{k\}) = (1 - p)^{k-1} p, \quad k \in \Omega,$$

is popularly called **Geometric(p) measure**

- $\Omega = \mathbb{N} \cup \{0\}$, $\mathcal{F} = 2^\Omega$.

Fix $\lambda \in (0, +\infty)$. The measure $\mathbb{P}$ defined by

$$\mathbb{P}(\{k\}) = \exp(-\lambda) \frac{\lambda^k}{k!}, \quad k \in \Omega,$$

is popularly called **Poisson(λ) measure**

Conditional Probabilities

Conditional Probability Measure

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Conditional Probability

Given $B \in \mathcal{F}$ such that $\mathbb{P}(B) > 0$, define $\mathbb{P}_B : \mathcal{F} \rightarrow [0, 1]$ as

$$\mathbb{P}_B(A) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \quad A \in \mathcal{F}.$$

Then, $\mathbb{P}_B$ is a valid probability measure on $(\Omega, \mathcal{F})$, and is called the **conditional probability measure** conditioned on the event B .

Notation: $\mathbb{P}_B(A)$ is denoted more commonly as $\mathbb{P}(A|B)$.

$\mathbb{P}_B$ is a Valid Probability Measure on $(\Omega, \mathcal{F})$

- **Proof that $\mathbb{P}_B(\emptyset) = 0$:**

$$\mathbb{P}_B(\emptyset) = \frac{\mathbb{P}(\emptyset \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(\emptyset)}{\mathbb{P}(B)} = 0.$$

- **Proof that $\mathbb{P}_B(\Omega) = 1$:**

$$\mathbb{P}_B(\Omega) = \frac{\mathbb{P}(\Omega \cap B)}{\mathbb{P}(B)} = \frac{\mathbb{P}(B)}{\mathbb{P}(B)} = 1.$$

- For any mutually disjoint collection of sets $A_1, A_2, \dots \in \mathcal{F}$,

$$\mathbb{P}_B\left(\bigsqcup_{k \in \mathbb{N}} A_k\right) = \sum_{k \in \mathbb{N}} \mathbb{P}_B(A_k).$$

Proof:

$$\mathbb{P}_B\left(\bigsqcup_{k \in \mathbb{N}} A_k\right) = \frac{\mathbb{P}\left(B \cap \bigsqcup_{k \in \mathbb{N}} A_k\right)}{\mathbb{P}(B)} = \frac{\mathbb{P}\left(\bigsqcup_{k \in \mathbb{N}} (B \cap A_k)\right)}{\mathbb{P}(B)} = \frac{\sum_{k \in \mathbb{N}} \mathbb{P}(B \cap A_k)}{\mathbb{P}(B)} = \sum_{k \in \mathbb{N}} \mathbb{P}_B(A_k).$$

Conditional Probability – Properties

- Fix $B \in \mathcal{F}$ such that $0 < \mathbb{P}(B) < 1$. Then, for any $A \in \mathcal{F}$,

$$\mathbb{P}(A) = \mathbb{P}(A|B) \cdot \mathbb{P}(B) + \mathbb{P}(A|B^c) \cdot \mathbb{P}(B^c).$$

Conditional Probability – Properties

- (Law of Total Probability)**

Suppose $B_1, B_2, \dots \in \mathcal{F}$ form a **partition** of Ω , i.e.,

$$B_i \cap B_j = \emptyset \quad \forall i \neq j, \quad \bigsqcup_{n \in \mathbb{N}} B_n = \Omega.$$

Then, for any $A \in \mathcal{F}$,

$$\mathbb{P}(A) = \sum_{n: \mathbb{P}(B_n) > 0} \mathbb{P}(A|B_n) \cdot \mathbb{P}(B_n).$$

Conditional Probability – Properties

- (Bayes' Theorem)**

Suppose $B_1, B_2, \dots \in \mathcal{F}$ form a **partition** of Ω , i.e.,

$$B_i \cap B_j = \emptyset \quad \forall i \neq j, \quad \bigsqcup_{n \in \mathbb{N}} B_n = \Omega.$$

For any $A \in \mathcal{F}$ such that $\mathbb{P}(A) > 0$,

$$\mathbb{P}(B_n|A) = \begin{cases} \frac{\mathbb{P}(A|B_n) \cdot \mathbb{P}(B_n)}{\sum_{j: \mathbb{P}(B_j) > 0} \mathbb{P}(A|B_j) \cdot \mathbb{P}(B_j)}, & \mathbb{P}(B_n) > 0, \\ 0, & \mathbb{P}(B_n) = 0. \end{cases}$$

Conditional Probability – Properties

- **(Chain Rule)**

Let $A_1, A_2, \dots \in \mathcal{F}$. Then,

$$\begin{aligned}\mathbb{P}\left(\bigcap_{n \in \mathbb{N}} A_n\right) &= \mathbb{P}(A_1) \cdot \mathbb{P}(A_2|A_1) \cdot \mathbb{P}(A_3|A_1 \cap A_2) \cdots \\ &= \mathbb{P}(A_1) \cdot \prod_{n \geq 2} \mathbb{P}\left(A_n \mid \bigcap_{j=1}^{n-1} A_j\right),\end{aligned}$$

provided each of the conditional probabilities on the right-hand side is defined.

Independence of Events

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Definition (Independence of Events)

Events $A, B \in \mathcal{F}$ are said to be **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \cdot \mathbb{P}(B).$$

We write $A \perp\!\!\!\perp B$ as a shorthand notation to denote that A and B are independent.

Remarks:

- The definition of independence does not involve conditional probabilities
- If $\mathbb{P}(B) > 0$, then

$$A \perp\!\!\!\perp B \iff \mathbb{P}(A|B) = \mathbb{P}(A).$$

- For $n > 2$, events $A_1, \dots, A_n$ are said to be independent if any subset of them are independent, i.e.,

$$\mathbb{P}\left(\bigcap_{i \in \mathcal{I}_0} A_i\right) = \prod_{i \in \mathcal{I}_0} \mathbb{P}(A_i) \quad \forall \mathcal{I}_0 \subseteq \{1, \dots, n\}.$$

Self-Independent Events

Self-Independent Events

Under what conditions is an event independent of itself?

- For A to be independent of itself,

$$\mathbb{P}(A \cap A) = \left(\mathbb{P}(A)\right)^2 \iff \mathbb{P}(A) \in \{0, 1\}.$$

- Thus, A is independent of itself if and only if its probability is either 0 or 1
- If $\mathbb{P}(A) \in \{0, 1\}$, then A is independent of every other event

Random Variables

Random Variables

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Random Variables)

A function $X : \Omega \rightarrow \mathbb{R}$ is called a **random variable** with respect to $\mathcal{F}$ if:

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F} \quad \forall x \in \mathbb{R}.$$

Remarks:

- A random variable is neither random nor a variable; it is a deterministic function
- The definition of a random variable is closely tied to the underlying σ -algebra $\mathcal{F}$
A function X that is a random variable with respect to $\mathcal{F}$ may not be a random variable with respect to $\mathcal{G}$ for $\mathcal{G} \neq \mathcal{F}$
- If X is a random variable with respect to $\mathcal{F}$, it is called an **$\mathcal{F}$ -measurable function**
- The definition of a random variable does not involve the underlying probability measure $\mathbb{P}$

Examples

- $\Omega = \{1, 2, \dots, 6\}$, $\mathcal{F} = \{\emptyset, \Omega\}$, $X(\omega) = \omega$
Is X a random variable with respect to $\mathcal{F}$?

Examples

- $\Omega = \{1, 2, \dots, 6\}, \quad \mathcal{F} = \left\{ \emptyset, \Omega \right\}, \quad X(\omega) = \omega$

Is X a random variable with respect to $\mathcal{F}$?

- What functions X are random variables with respect to $\mathcal{F}$?

Examples

- $\Omega = \{1, 2, 3, 4, 5\}$, $\mathcal{F} = \left\{ \emptyset, \Omega, A, A^c \right\}$ for a fixed $A \subseteq \Omega$

What functions X are random variables with respect to $\mathcal{F}$?

Examples

- $\Omega = \{1, 2, 3, 4, 5\}, \quad \mathcal{F} = \sigma \left(\left\{ \{1\}, \{2, 3\} \right\} \right)$

What functions X are random variables with respect to $\mathcal{F}$?

Examples

- $\Omega = \mathbb{N}$, $\mathcal{F} = 2^\Omega$

What functions X are random variables with respect to $\mathcal{F}$?